

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$x^2 + 6x + 4$

Which of the following is equivalent to the expression above?

- Answer
- A. $(x + 3)^2 + 5$
- B. $(x + 3)^2 - 5$
- C. $(x - 3)^2 + 5$
- D. $(x - 3)^2 - 5$

Correct Answer: B

Rationale

Choice B is correct. The given quadratic expression is in standard form, and each answer choice is in vertex form. Completing the square converts the expression from standard form to vertex form. The first step is to rewrite the expression as follows:

$x^2 + 6x + 4 = x^2 + 6x + 9 + 4 - 9$. The first three terms of the revised expression can be rewritten as a perfect square as follows:

$x^2 + 6x + 9 + 4 - 9 = (x + 3)^2 + 4 - 9$. Combining the constant terms gives $(x + 3)^2 - 5$.

Choice A is incorrect. Squaring the binomial and simplifying the expression in choice A gives $x^2 + 6x + 9 + 5$. Combining like terms gives $x^2 + 6x + 14$, not $x^2 + 6x + 4$. Choice C is incorrect. Squaring the binomial and simplifying the expression in choice C gives $x^2 - 6x + 9 + 5$. Combining like terms gives $x^2 - 6x + 14$, not $x^2 + 6x + 4$. Choice D is incorrect. Squaring the binomial and simplifying the expression in choice D gives $x^2 - 6x + 9 - 5$. Combining like terms gives $x^2 - 6x + 4$, not $x^2 + 6x + 4$.